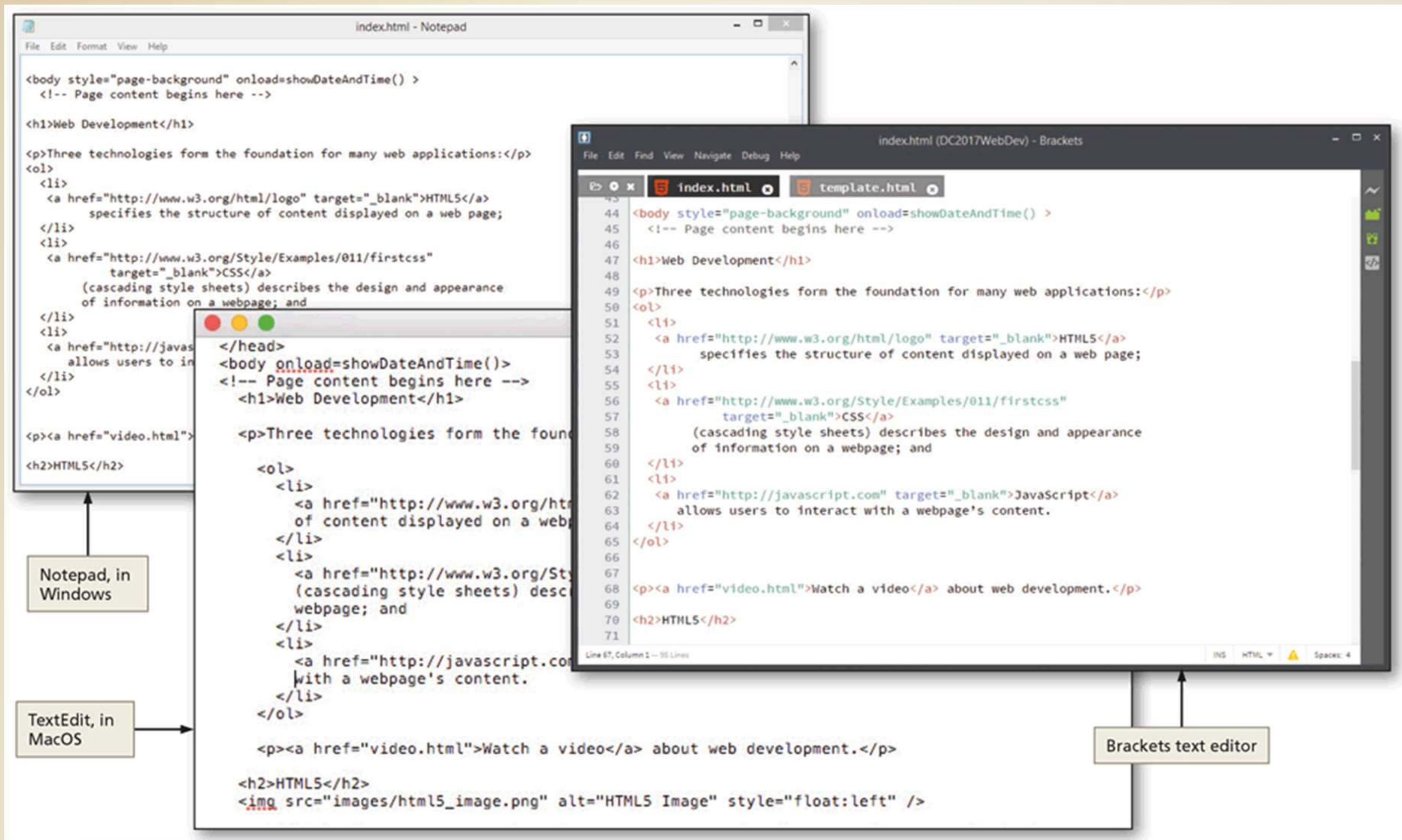
Review Web Technology

Tools for Creating a Website

- Text editors
  - Similar to a word processing program, but lacks most text formatting features
  - Operating systems typically include a text editor
  - A code editor is a type of text editor that contains additional features to help web developers write code

Tools for Creating a Website

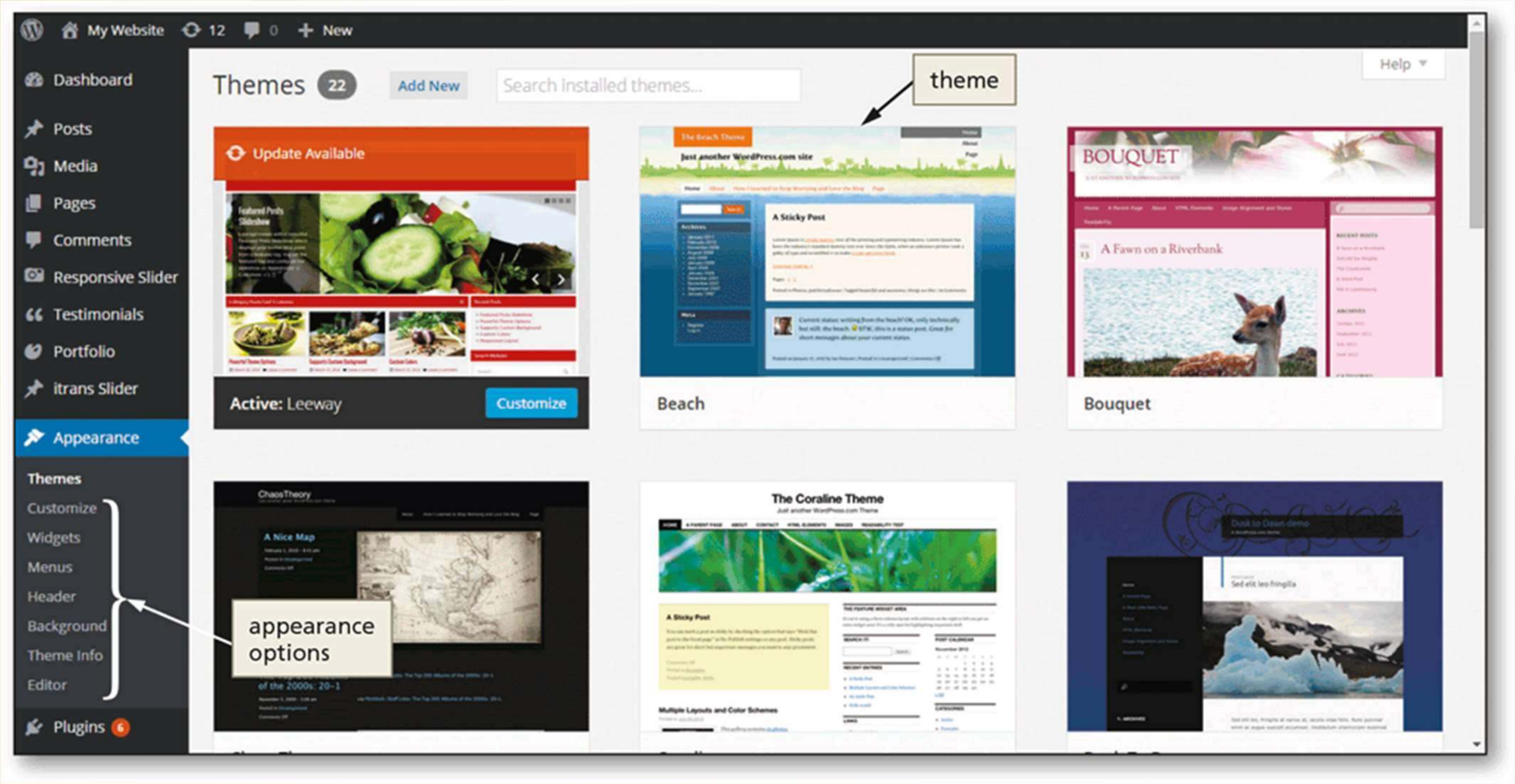


Page FO 2
Figure 1

Tools for Creating a Website

- Content Management Systems
  - Enables and manages the publishing, modification, organization, and access of various forms of documents and other files on a network or the web
  - Web developer creates a theme, and one or more website content administrators enters the content

Tools for Creating a Website

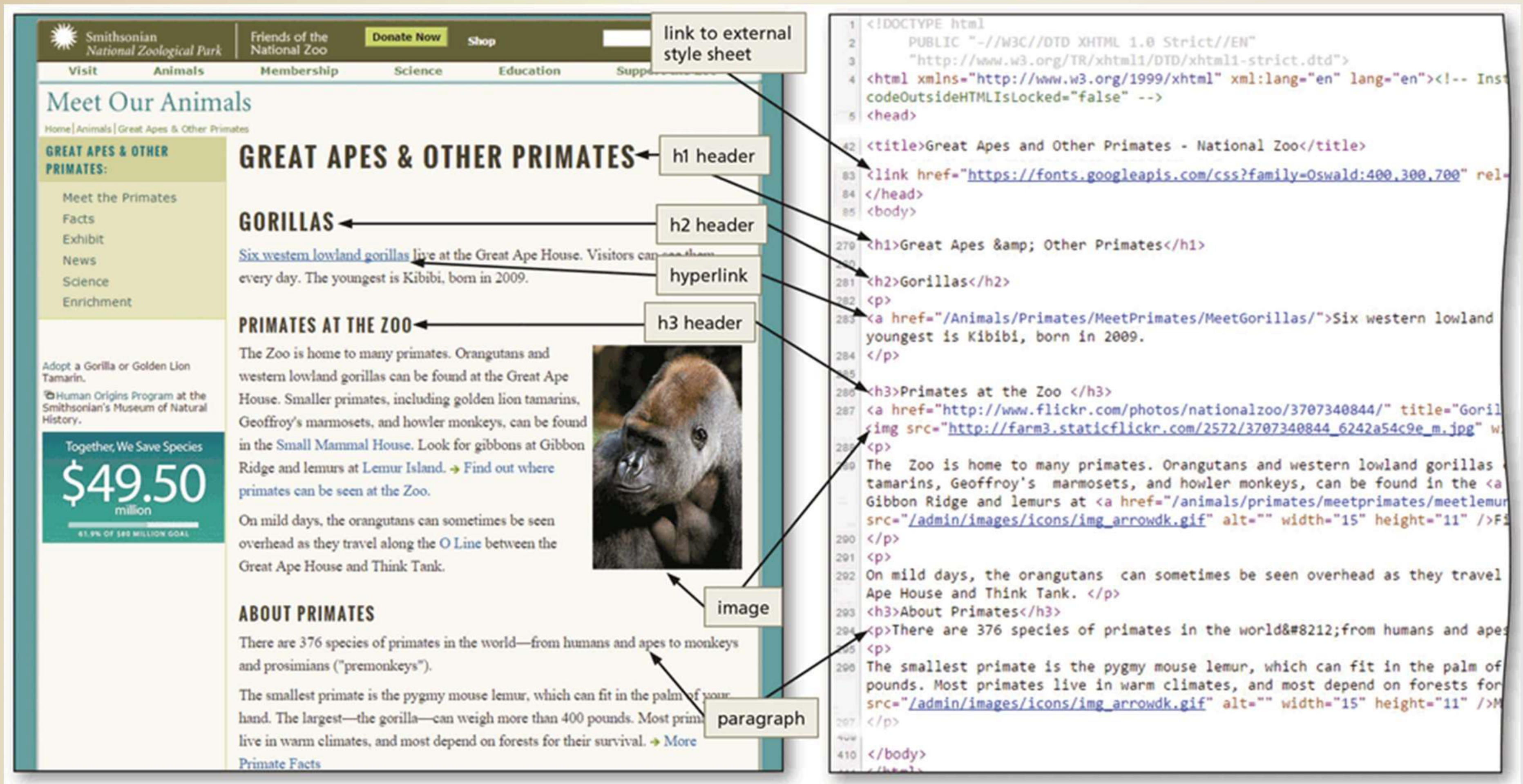


Page FO 4
Figure 2

Website Technologies

- Hypertext Markup Language (HTML) uses a set of codes called tags to format documents for display in a browser
- A complementary technology called cascading style sheets (CSS) contains specifications for the fonts, colors, layout, and placement of these HTML elements on a webpage
- JavaScript is a programming language for creating programs that a browser can run to generate content for a website

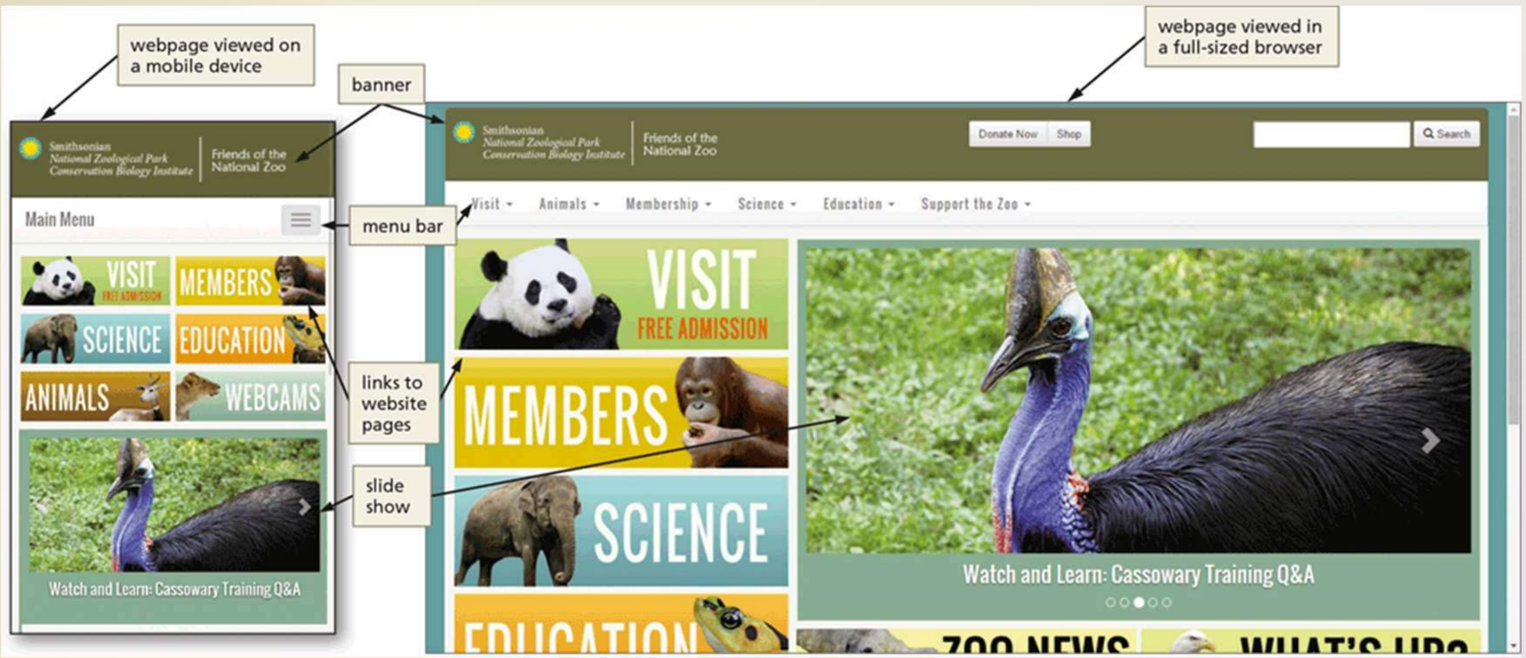
Website Technologies



Page FO 5
Figure 3

Website Technologies

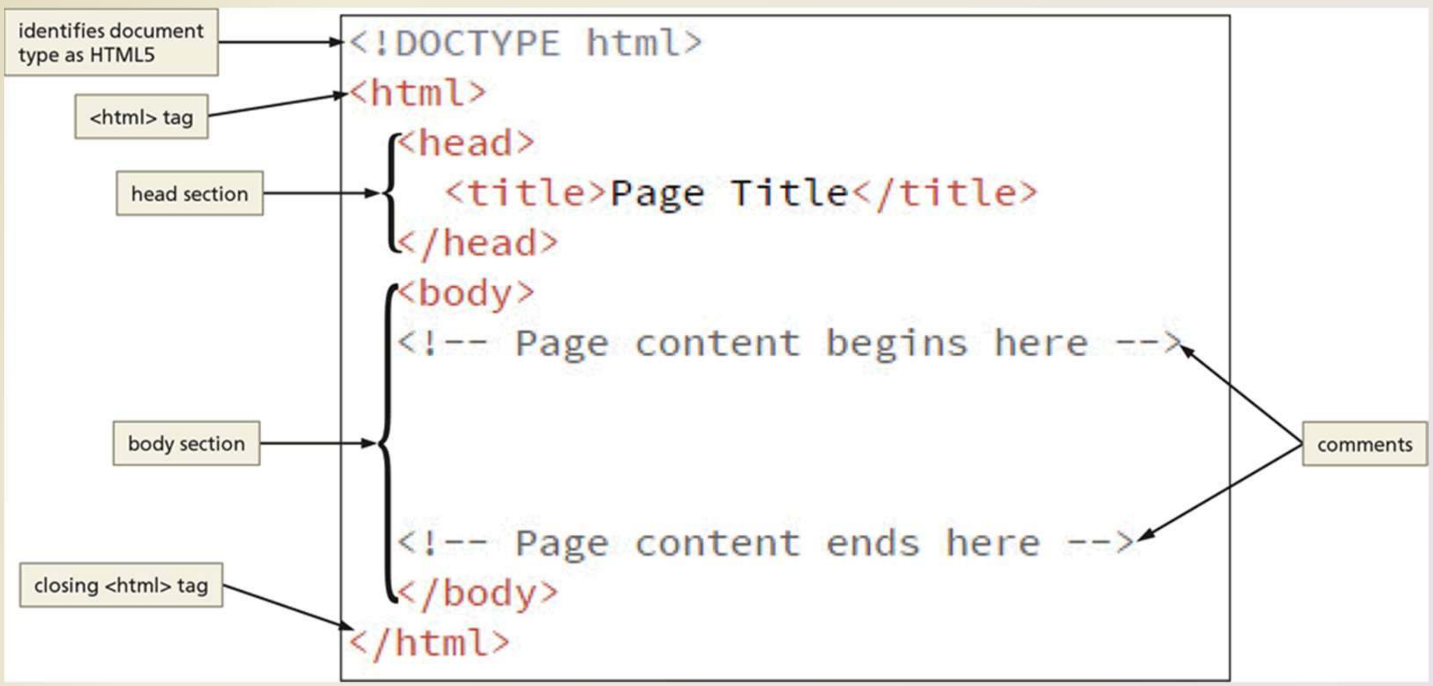
- Responsive webpages automatically adjust the size of their content to display appropriately relative to the size of the screen of the device on which it is displayed



Page FO 5
Figure 4

Structure of a Webpage

- A webpage’s source code contains text marked up with HTML tags that instruct a browser how to display that content



Page FO 6
Figure 5

Structure of a Webpage

- The World Wide Web Consortium (W3C) oversees the specification of HTML Standards
- The W3C provides a free, online HTML5 validator application to ensure that a webpage's HTML tags follow the specifications, or rules, for HTML5

Creating the index.html File

- Run the text editor of your choice
- Navigate to and open the template.html file
- If necessary, enable the word wrap feature so that you can view all the webpage text without scrolling horizontally
- Save the file using the file name, index.html. Do not exit the text editor

Creating the index.html File

```
<!DOCTYPE html>
<html>
  <head>
    <title>Page Title</title>
  </head>
  <body>
  <!-- Page content begins here -->



  <!-- Page content ends here -->
  </body>
</html>
```

webpage structure

Page FO 9
Figure 8

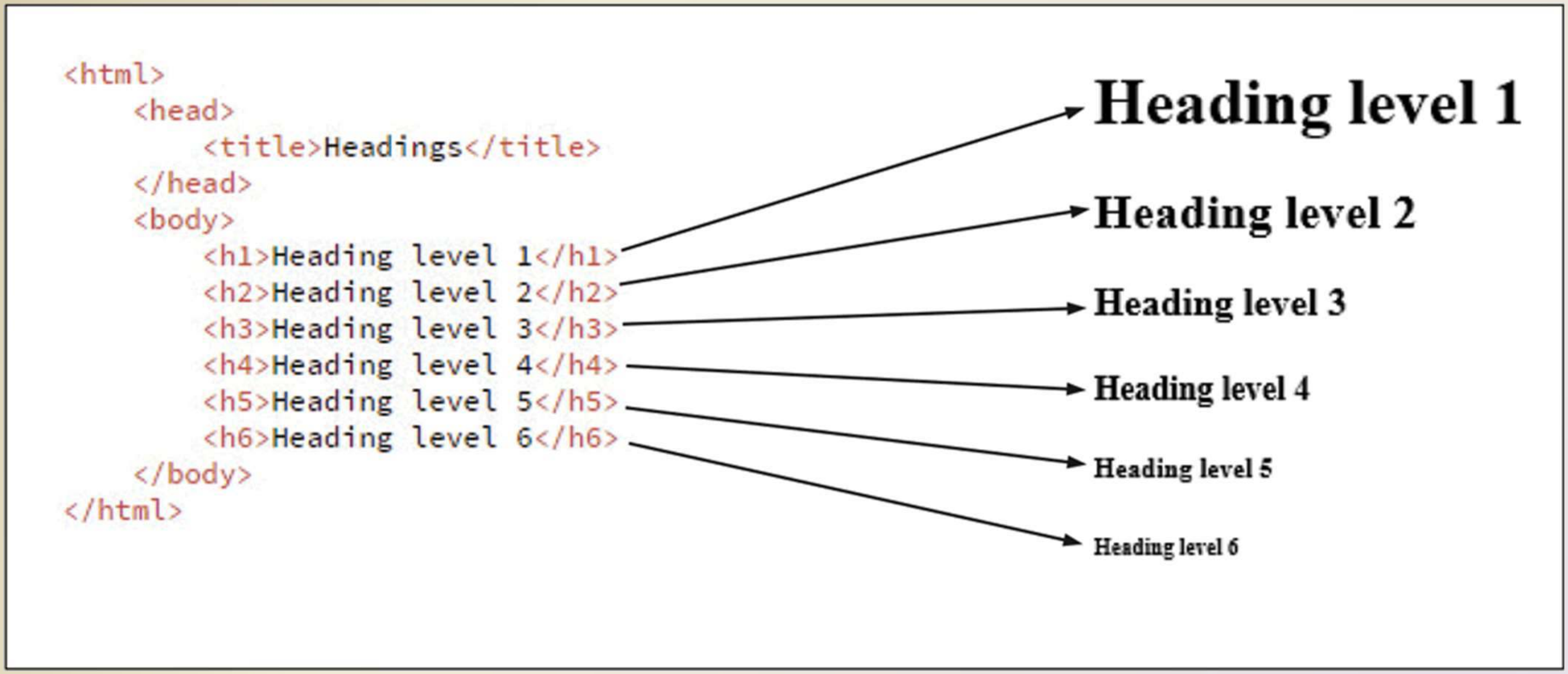
Adding the Webpage Title

- Select the text, Page Title, that appears between the `<title>` and `</title>` tags
- Type `Mark's Web Development Page` as the title. Replace the name, Mark, with your first name

Headings

- Headings indicate the different sections of a webpage. HTML supports six levels of headings, which are identified by the following tags: `<h1>`, `<h2>`, `<h3>`, `<h4>`, `<h5>`, and `<h6>`

Headings



Page FO 12
Figure 14

Paragraphs

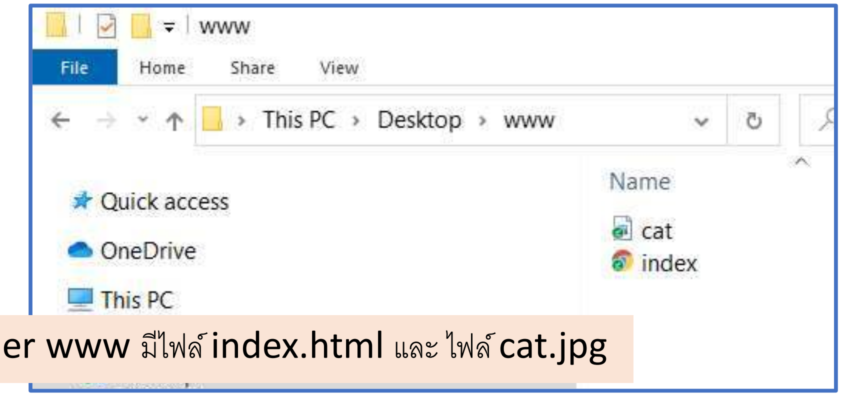
- The `<p>` and `</p>` tags are used to identify the beginning and ending of Paragraphs
- If you have several paragraphs of text on your webpage, these tags will inform the browser to insert additional line spacing above and below the paragraph so that the text is easier to read when displayed in the browser
  - The browser ignores line breaks and line spacing in the HTML file, so it is important to properly define the paragraphs using the `<p>` and `</p>` tags

Images

- Images can be either photos or graphics
- Images always are stored as separate files, and references to the images appear in the HTML code using the `<img>` tag
- Common attributes for the `<img>` tag describe the location of an image file, alternate text for the image, and a style that indicates how to position the image

การอ้างอิงตำแหน่งไฟล์รูป ใน html

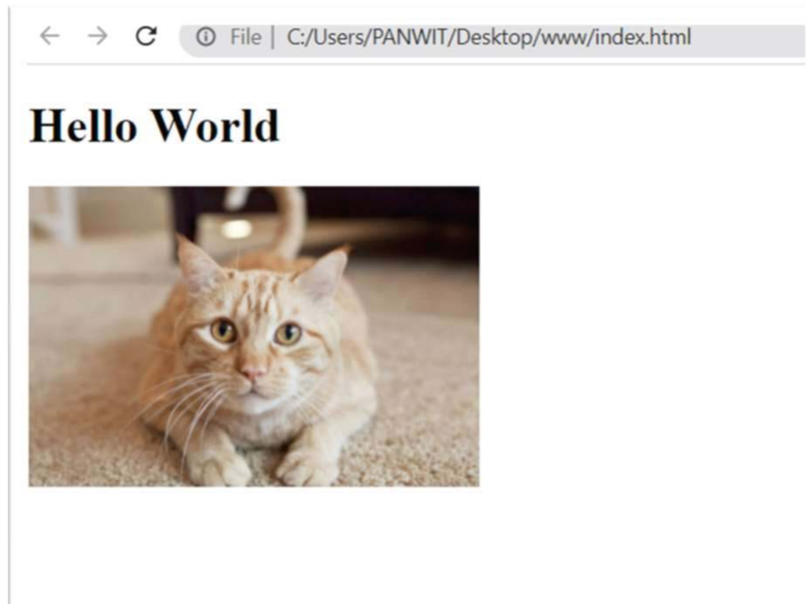
- File นั้นๆ อยู่ใน folder เดียวกัน
  เช่น ไฟล์รูป และไฟล์ index.html
  อยู่ใน folder เดียวกัน



ใน folder www มีไฟล์ index.html และ ไฟล์ cat.jpg

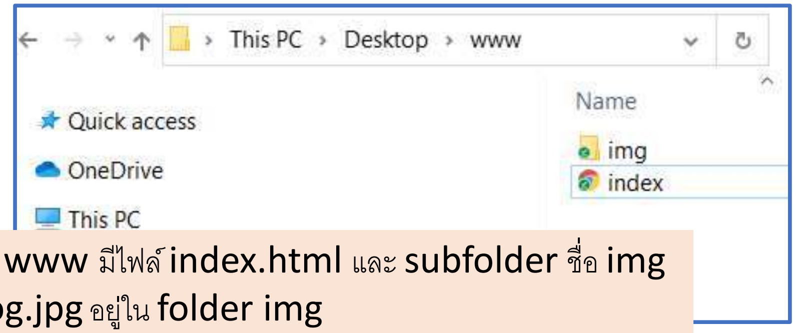
ไฟล์ index.html

```
<!DOCTYPE html>
<html>
    <head>
        <title>My Website Title</title>
    </head>
    <body>
        <h1>Hello World</h1>
        <img src="cat.jpg" height="200" width="300">
    </body>
</html>
```

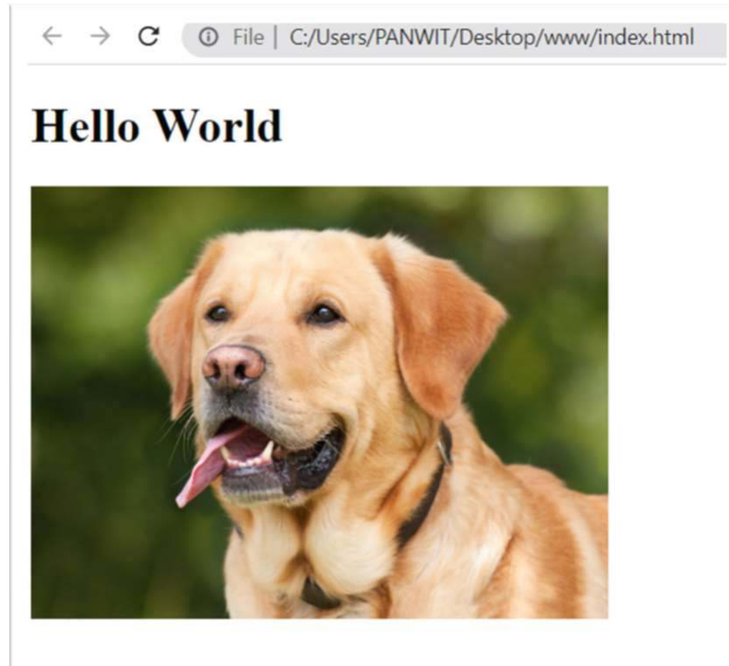


การอ้างอิงตำแหน่งไฟล์รูป ใน html

- File นั้นๆ อยู่คนละ folder
  เช่น ไฟล์รูป และไฟล์ index.html
  อยู่คนละ folder



ใน folder www มีไฟล์ index.html และ subfolder ชื่อ img
และมีรูป dog.jpg อยู่ใน folder img



ไฟล์ index.html

```
<!DOCTYPE html>
<html>
     <head>
          <title>My Website Title</title>
     </head>
     <body>
          <h1>Hello World</h1>
          <img src="img/dog.jpg" height="288" width="384"
>
     </body>
</html>
```

Links

- A link, or hyperlink, can be text or an image in a webpage that a user clicks to navigate to another webpage, download a file, or perform another action, such as running an email app and addressing an email message

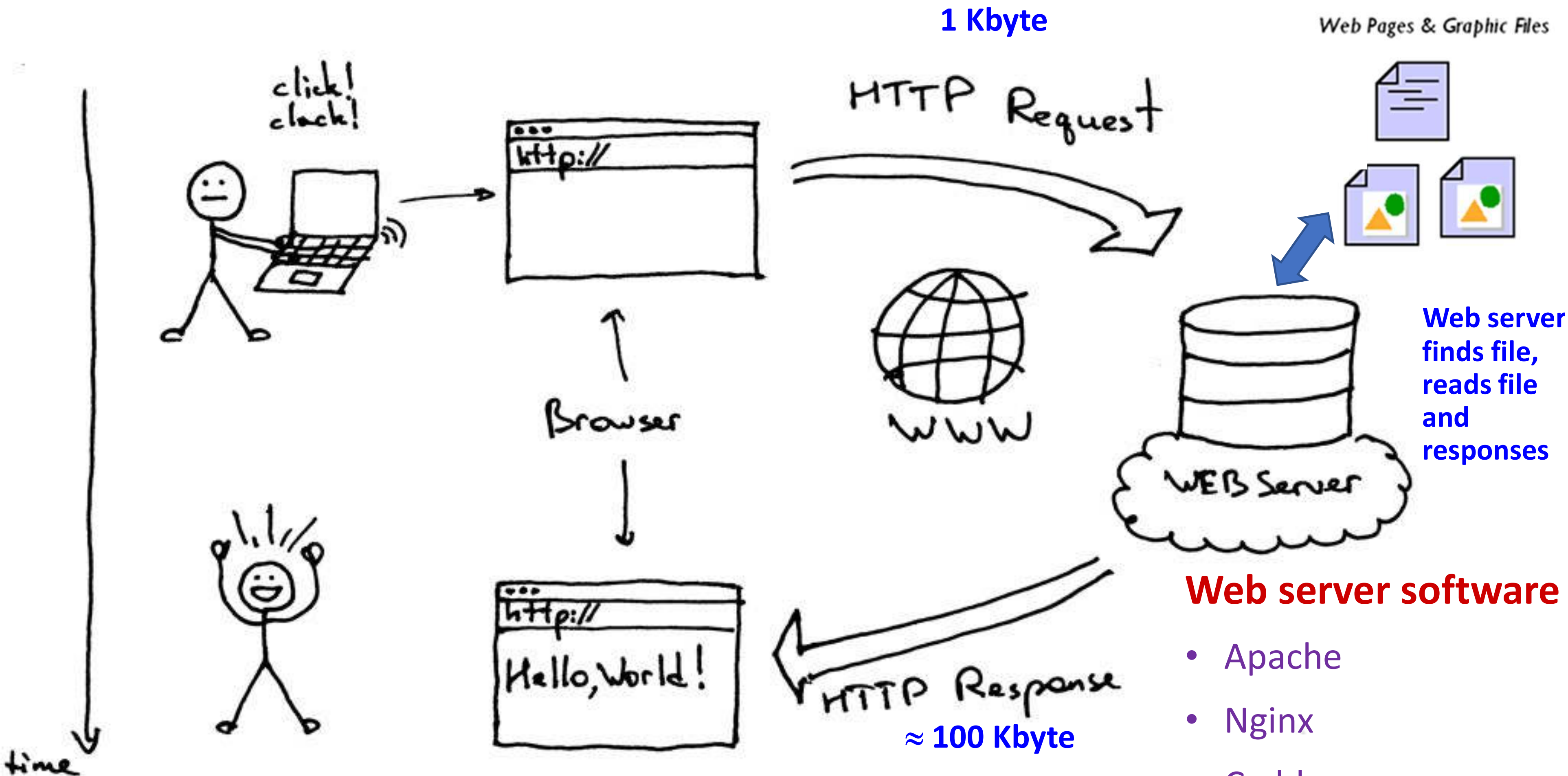
HYPERLINK WITH ABSOLUTE REFERENCE

```
<a href="http://www.w3.org" target="_blank">HTML5</a>
```

HYPERLINK WITH RELATIVE REFERENCE

```
<a href="video.html"> Watch a video</a>
```

1. ภาพรวม web client-server



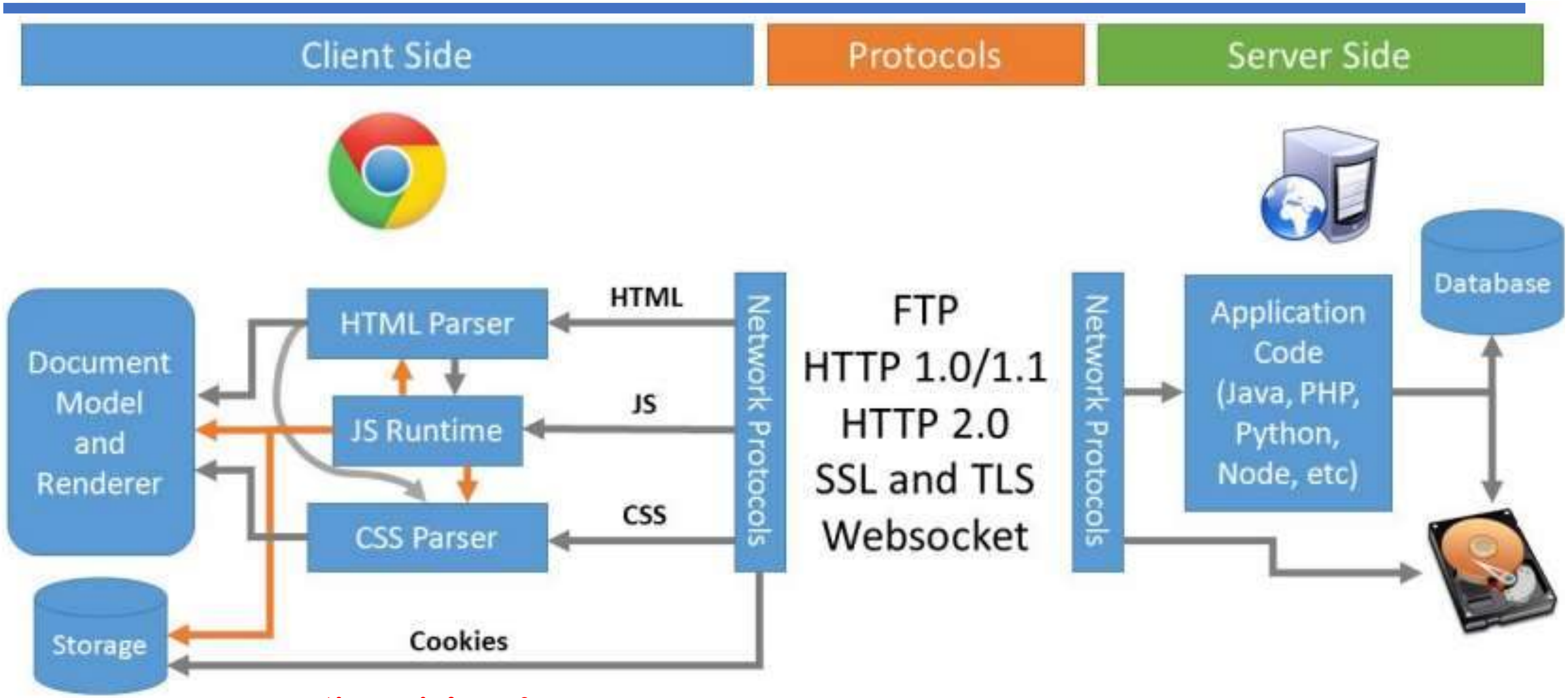
Web server software

- Apache
- Nginx
- Caddy
- Microsoft IIS

Web Browser Software:

- Chrome, Safari, Firefox, Microsoft Edge

2. การทำงานของ Web Browser

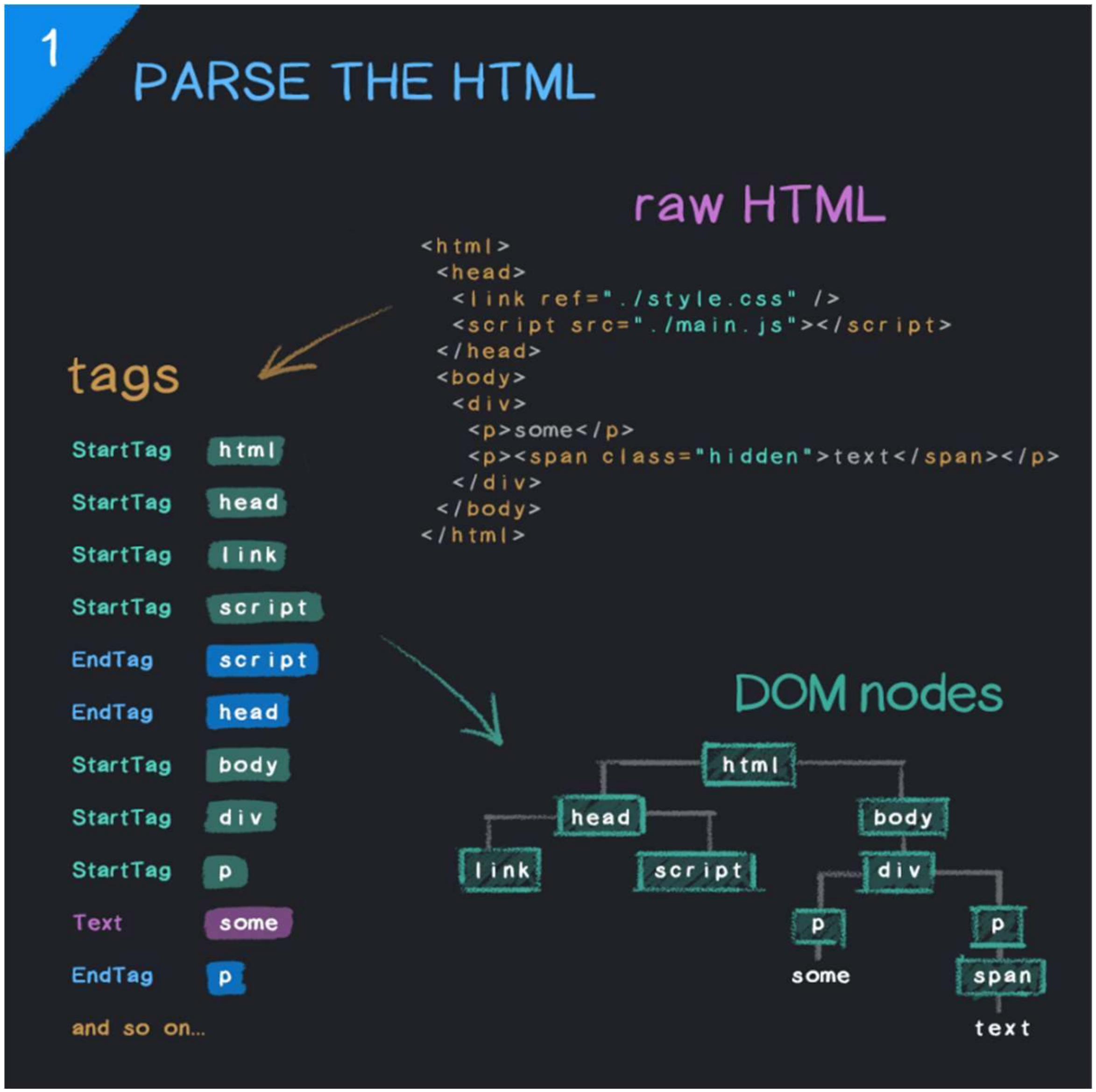


ตัววิเคราะห์คำ แยกคำ ในประโยค

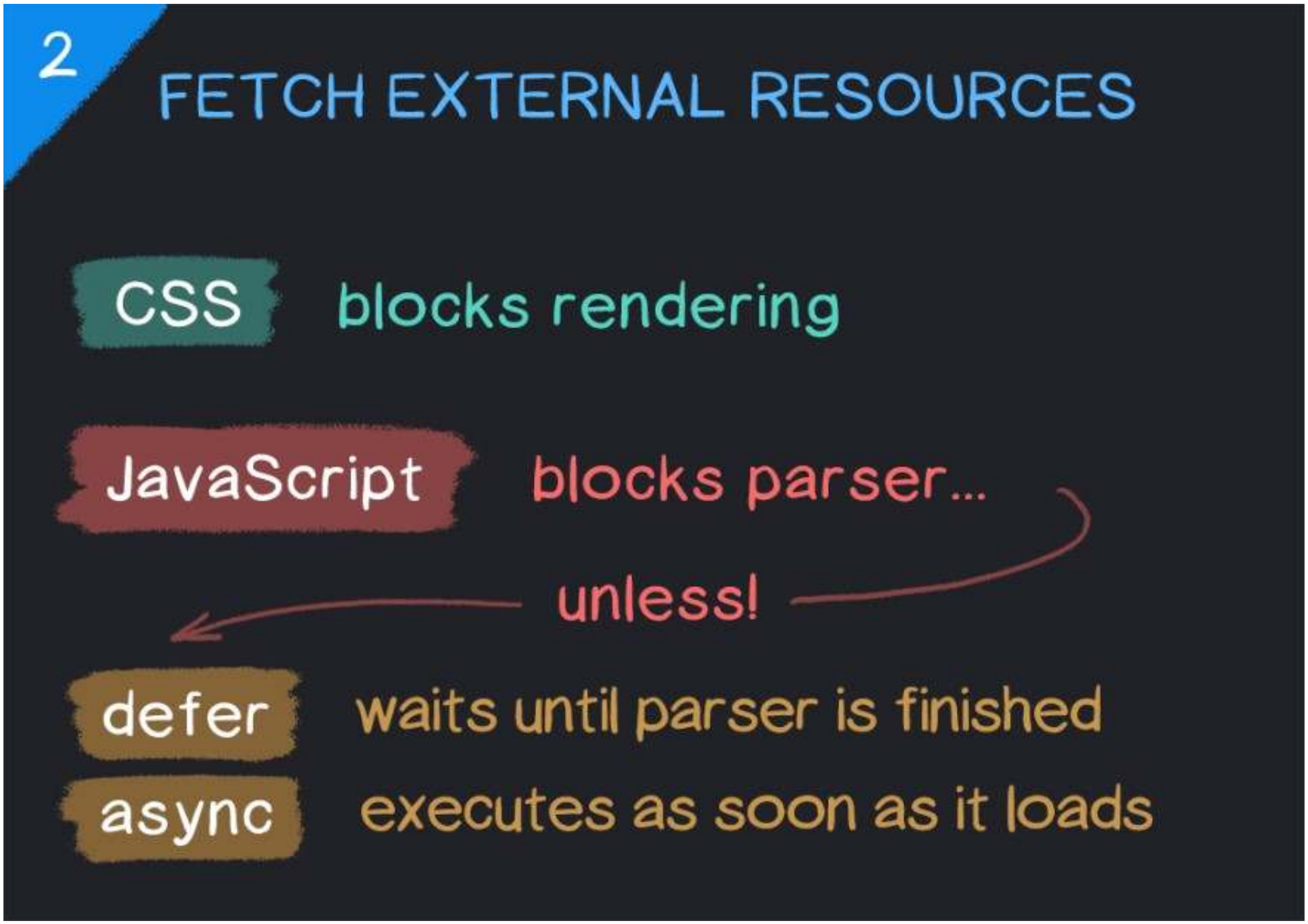
- HTML Parser : ทำหน้าที่ render พวก tag ต่าง ๆ ของ code html ออกมาเป็น DOM
- CSS Parser : ส่วนไหนใน code html เป็น css ก็จะเข้า CSS Parser ออกมาเป็น CSSOM
- JS Runtime : ถ้าใน code html มี JavaScript (js) มันก็จะต้องการ JS runtime environment ในการ run ให้ code js ทำงาน โดยสามารถใช้ js ทำการ อ่าน/แก้ไข html หรือจัดการ css ได้ ซึ่ง browser ทุกตัวบนโลกเดี๋ยวนี้มี js runtime อยู่แล้ว

JavaScript Runtime ทำหน้าเพื่อ อ่าน code ภาษา java script และทำงานตามคำสั่ง เพื่อให้ os ต่างๆ ทำงานได้

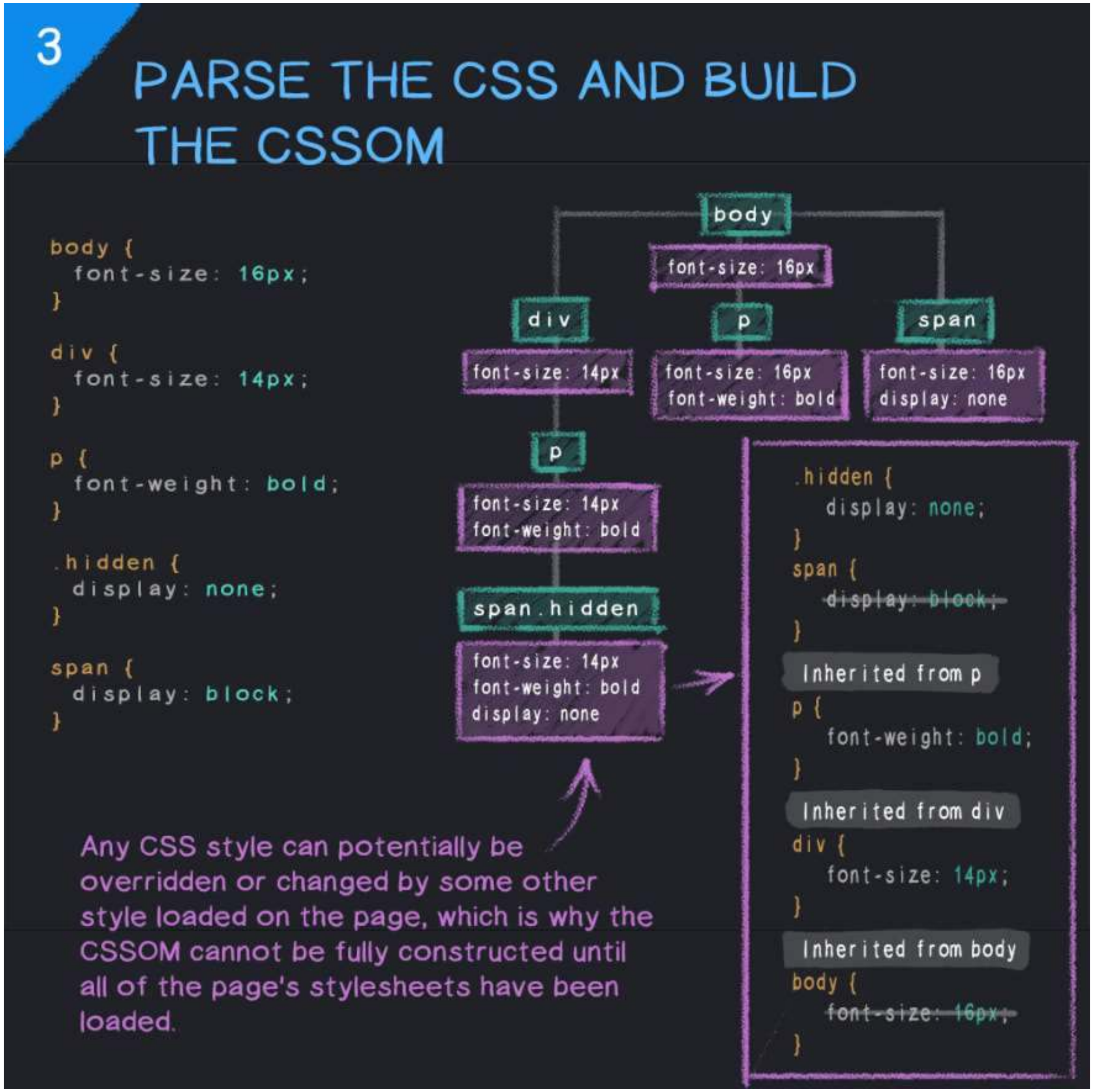
2. การทำงานของ Web Browser



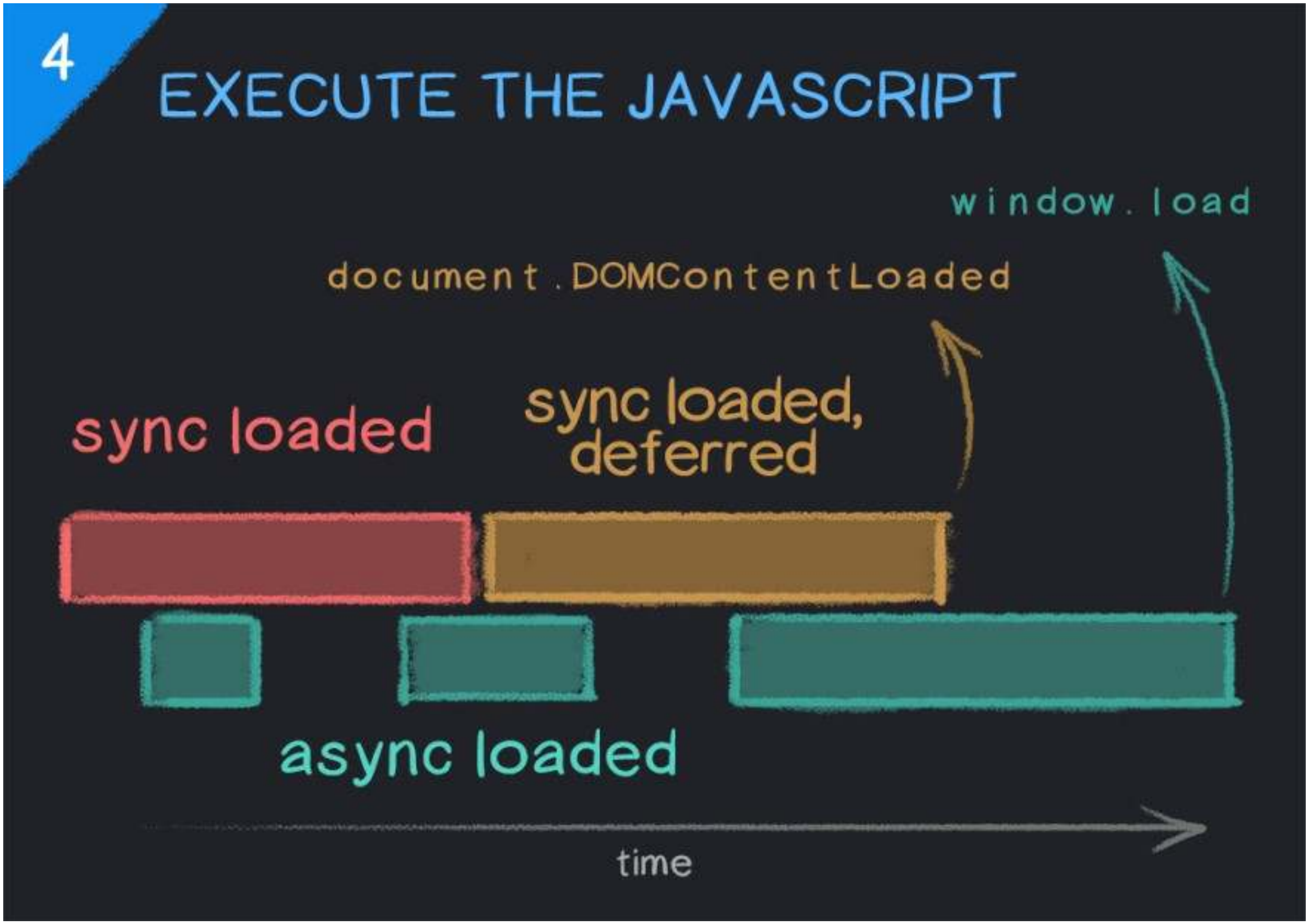
2. การทำงานของ Web Browser



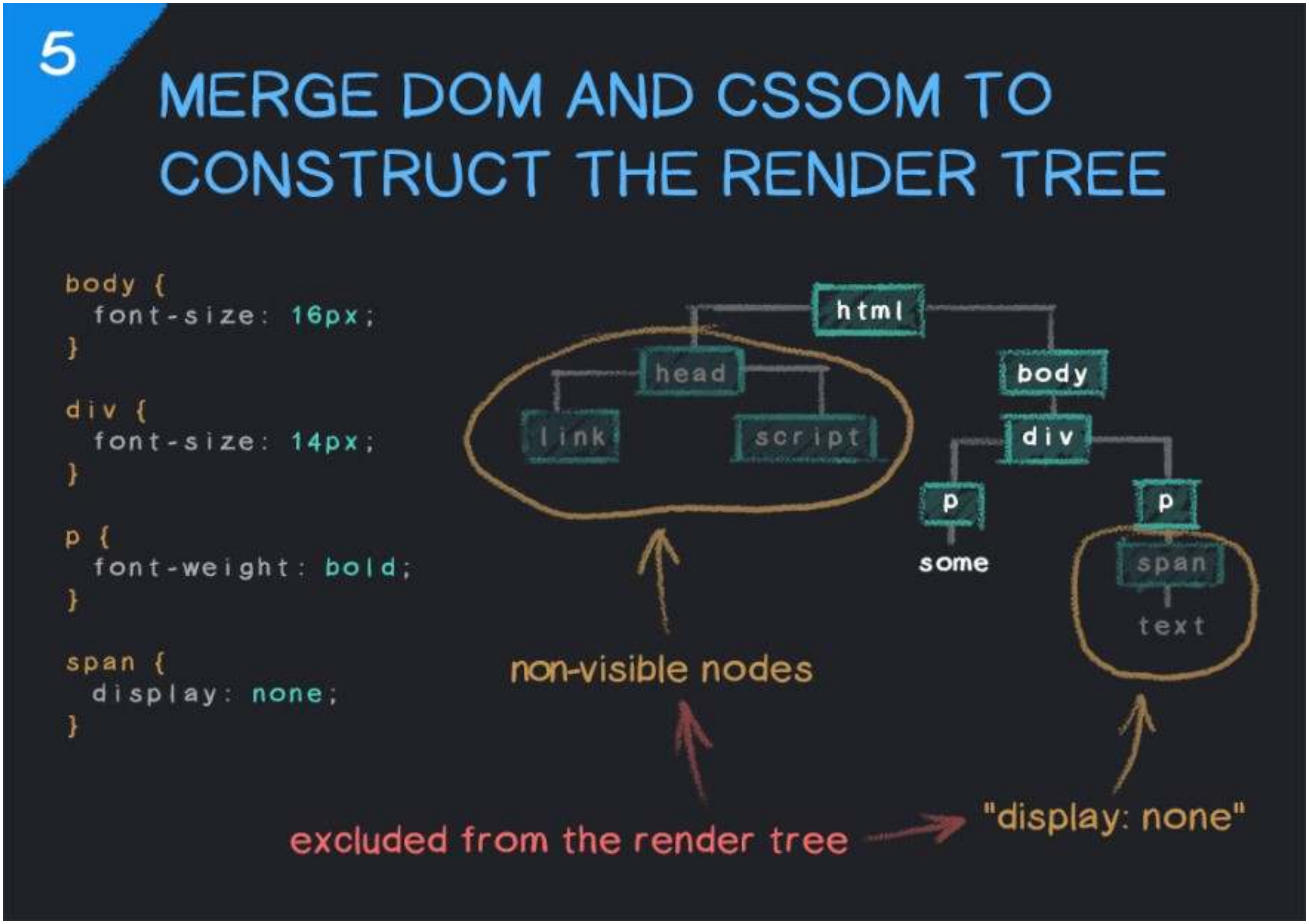
2. การทำงานของ Web Browser



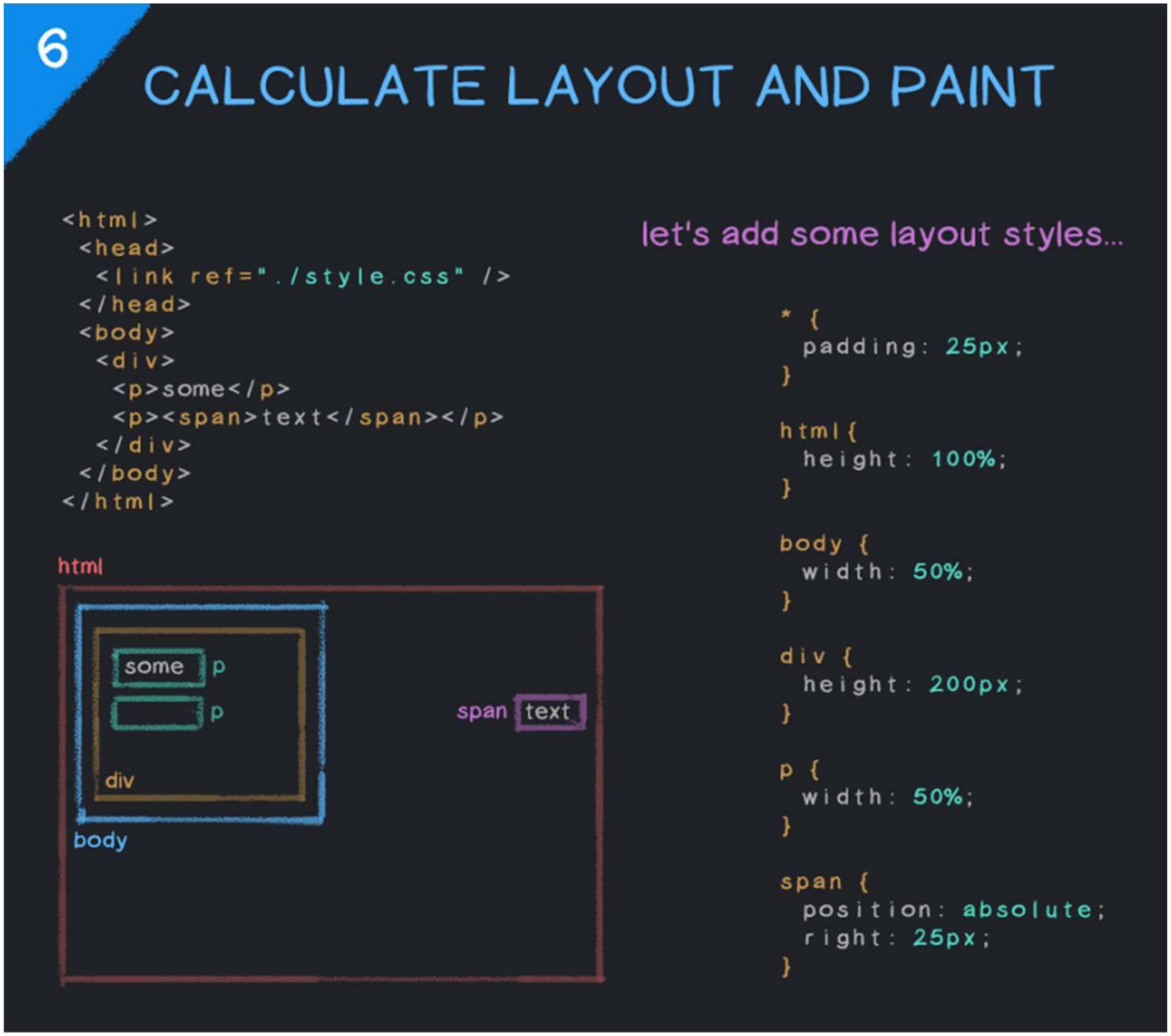
2. การทำงานของ Web Browser



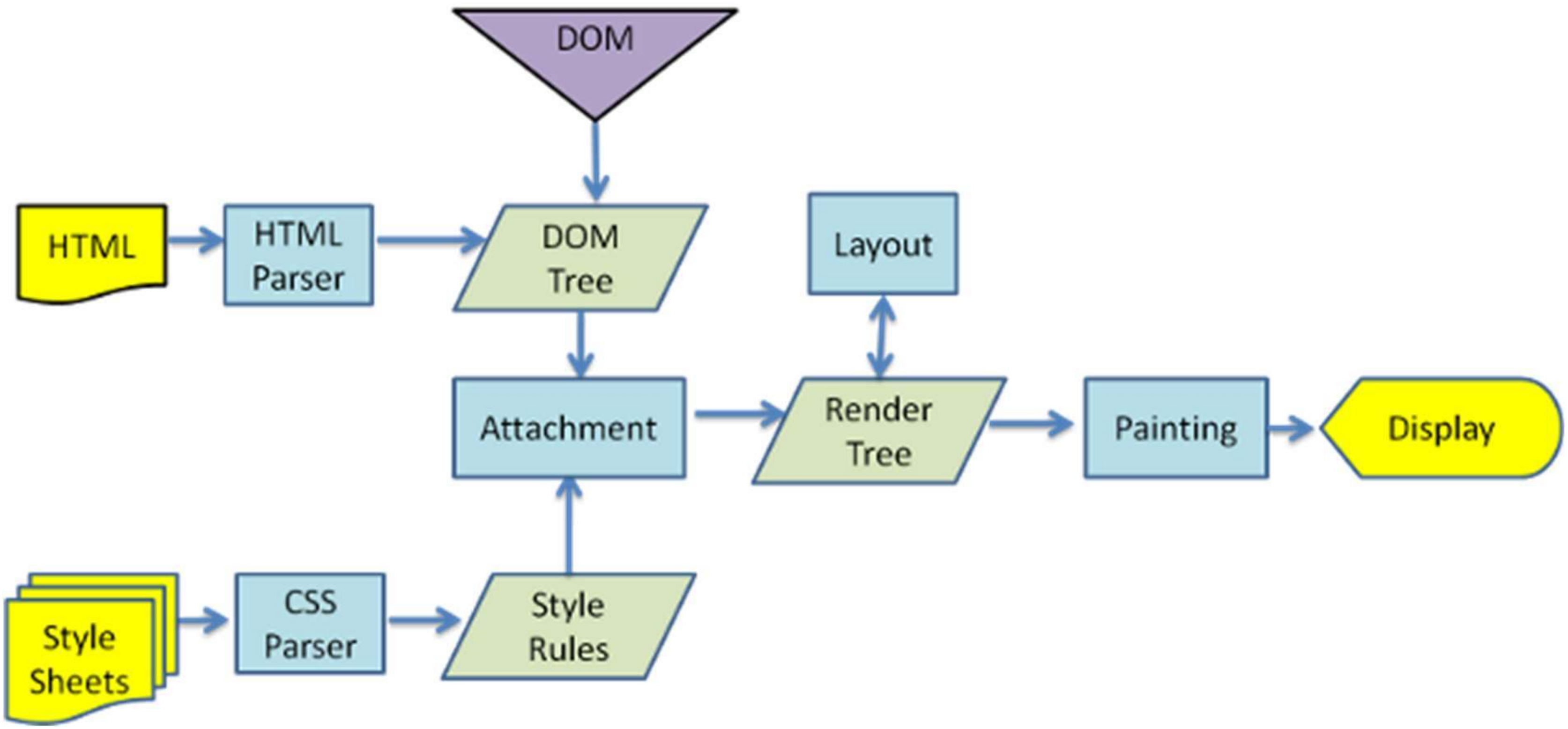
2. การทำงานของ Web Browser



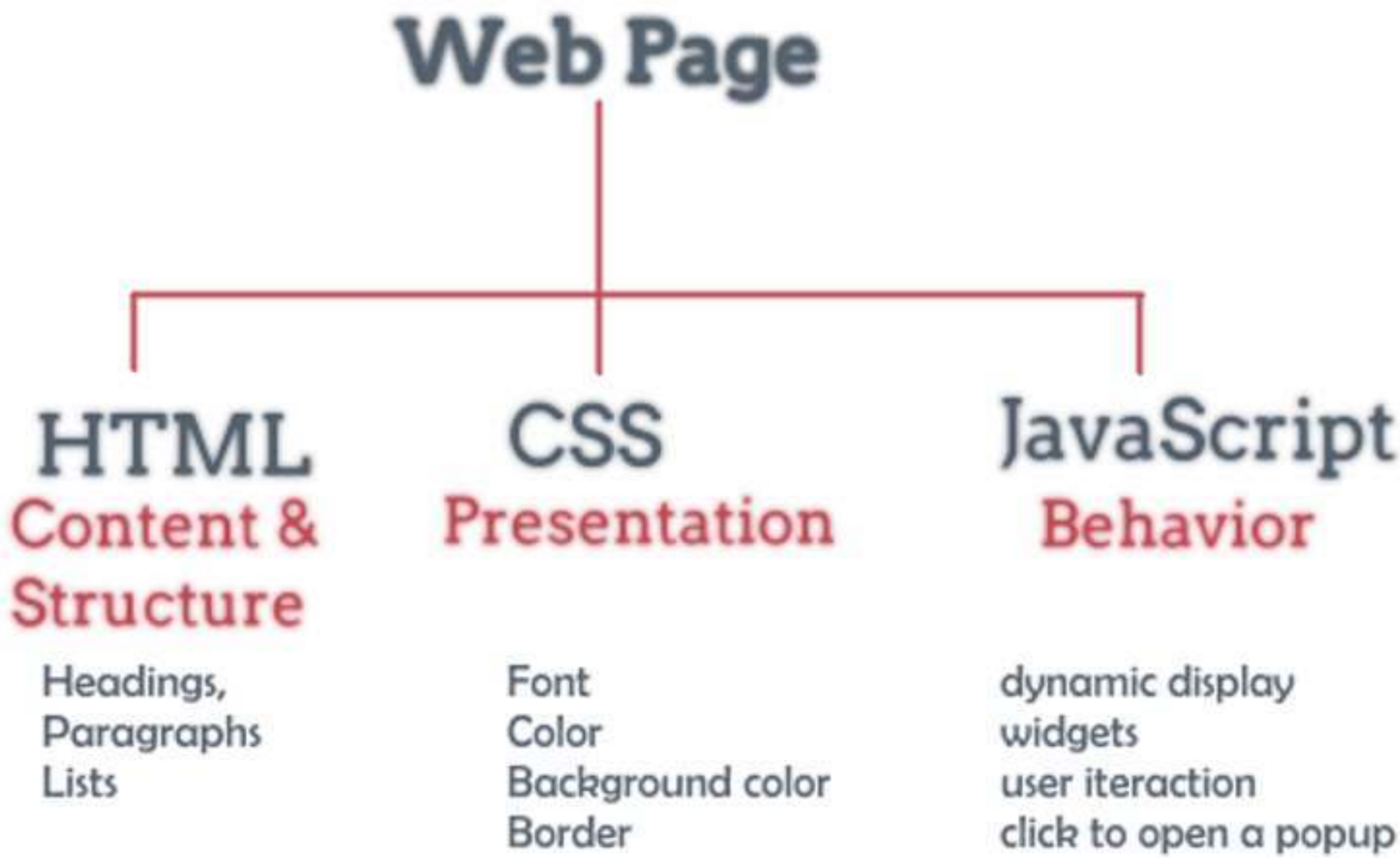
2. การทำงานของ Web Browser



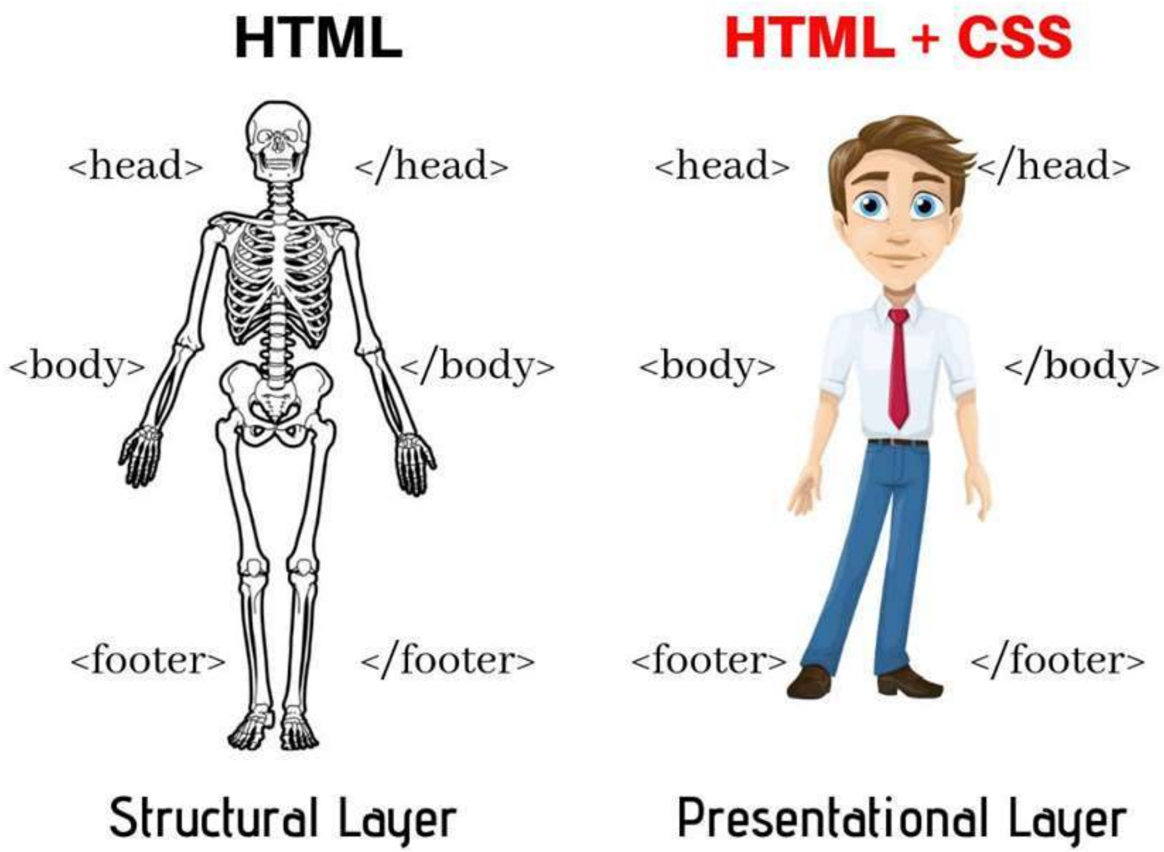
2. การทำงานของ Web Browser



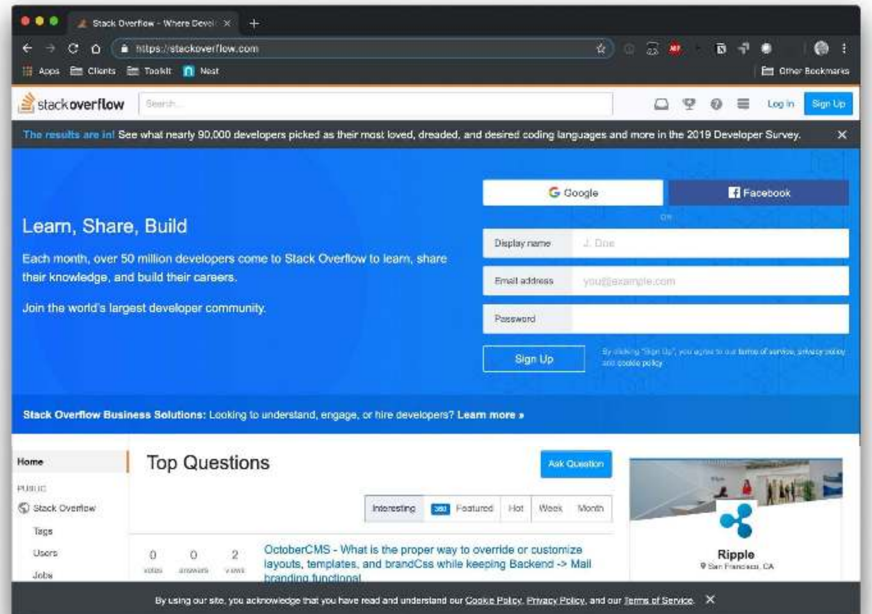
3. Html CSS JavaScript

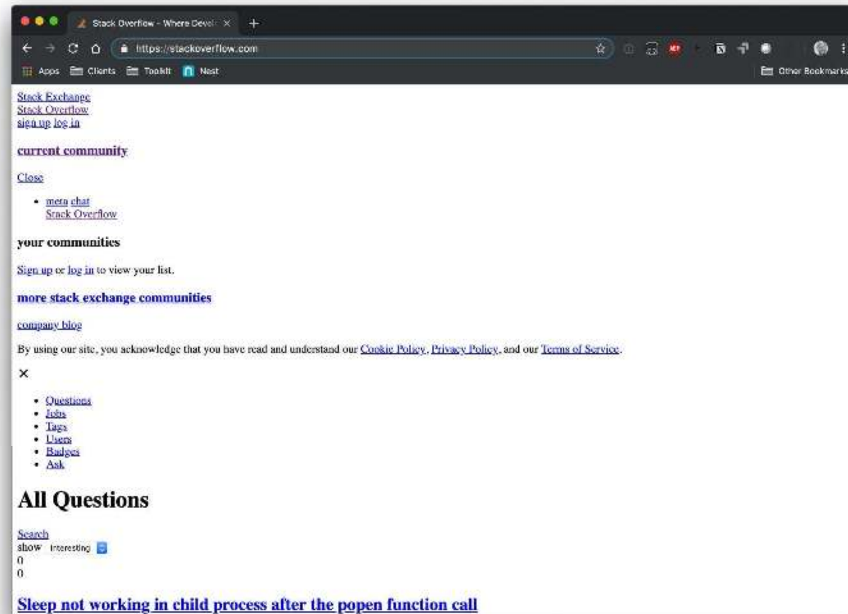


HTML Vs CSS


HTML
HTML + CSS
<head>
</head>
<body>
</body>
<footer>
</footer>
<head>
</head>
<body>
</body>
<footer>
</footer>
Structural Layer
Presentational Layer

HTML Vs CSS


Stack Overflow - Where Devel
https://stackoverflow.com
Apps
Clients
Toolkit
Nest
Other Bookmarks
stackoverflow
Log in
Sign Up
The results are in! See what nearly 90,000 developers picked as their most loved, dreaded, and desired coding languages and more in the 2019 Developer Survey.
Learn, Share, Build
Each month, over 50 million developers come to Stack Overflow to learn, share their knowledge, and build their careers.
Join the world's largest developer community.
Google
Facebook
Display name
Email address
Password
Sign Up
Stack Overflow Business Solutions: Looking to understand, engage, or hire developers? Learn more »
Home
Top Questions
Ask Question
Stack Overflow
Tags
Users
Jobs
Interesting
Featured
Hot
Week
Month
OctoberCMS - What is the proper way to override or customize layouts, templates, and brandCss while keeping Backend -> Mail branding functional
Ripple
By using our site, you acknowledge that you have read and understand our Cookie Policy, Privacy Policy, and our Terms of Service.


Stack Overflow - Where Devel
https://stackoverflow.com
Apps
Clients
Toolkit
Nest
Other Bookmarks
Stack Exchange
Stack Overflow
sign up log in
current community
Close
meta chat
Stack Overflow
your communities
Sign up or log in to view your list.
more stack exchange communities
company blog
By using our site, you acknowledge that you have read and understand our Cookie Policy, Privacy Policy, and our Terms of Service.
Questions
Jobs
Tags
Users
Badges
Ask
All Questions
Search
show
Sleep not working in child process after the popen function call

3. Applying Styles with CSS

- Although it is possible to customize these elements of webpages using HTML, CSS makes it easier to specify the appearance of similar elements in the same webpage or same website
- Embedded styles, which define styles in the **`<head>`** section of the index.html document, apply to the entire webpage on which they are defined

```
<style>
    h1 {
      font-family:sans-serif;
      color:navy;
      font-style:italic;
    }
    h2 {
      font-family:cursive;
      background-color: navy;
      color:papayawhip;
    }
    p {
      font-family:sans-serif;
      color:rgb(56,0,0);
    }
    ol {
      font-family:sans-serif;
      color:rgb(56,0,0);
    }
    body {
      background-color:#bbccff;
    }
    .fancy {
      font-weight:bold;
      color:red;
      font-style:italic;
    }
</style>
```

Adding CSS to the index.html File

styles entered in head section of index.html

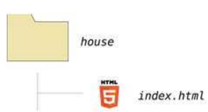
```
<!DOCTYPE html>
<html>
  <head>
    <title>Mark's Web Development Page</title>
    <style>

      h1 {
        font-family:sans-serif;
        color:navy;
        font-style:italic;
      }
      h2 {
        font-family:cursive;
        background-color:navy;
        color:papayawhip;
      }
      p {
        font-family:sans-serif;
        color:rgb(56,0,0);
      }
      ol {
        font-family:sans-serif;
        color:rgb(56,0,0);
      }
      body {
        background-color:#bbccff;
      }
      .fancy {
        font-weight:bold;
        color:red;
        font-style:italic;
      }
    </style>

  </head>
  <body>
  <!-- Page content begins here -->
    <h1>Web Development</h1>

    <p>Three technologies form the foundation for many web applications:</p>

      <ol>
```

Styling HTML with CSS

- CSS can be added to HTML elements in 3 ways
  - Inline - by using the style attribute in HTML elements
  - Internal - by using a <style> element in the <head> section
  - External - by using an external CSS file

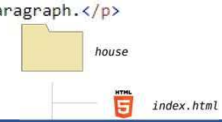
Inline

```html
<h1 style="color:blue;">This is a Blue Heading</h1>
```



Internal

```html
<!DOCTYPE html>
<html>
<head>
<style>
body {background-color: powderblue;}
h1   {color: blue;}
p    {color: red;}
</style>
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```



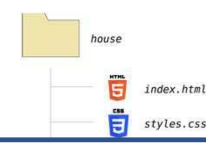
External

```html
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="styles.css">
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

ไฟล์ index.html

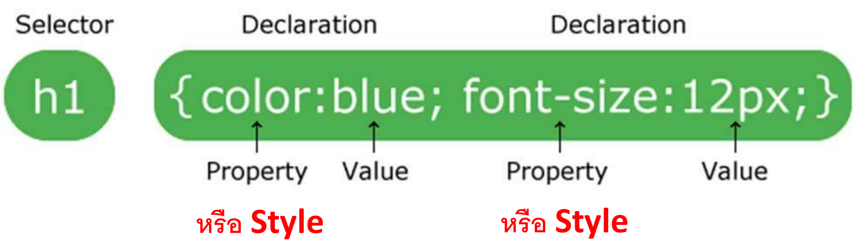
```css
body {
  background-color: powderblue;
}
h1 {
  color: blue;
}
p {
  color: red;
}
```

ไฟล์ style.css



3. CSS Syntax

A CSS rule-set consists of a selector and a declaration block:

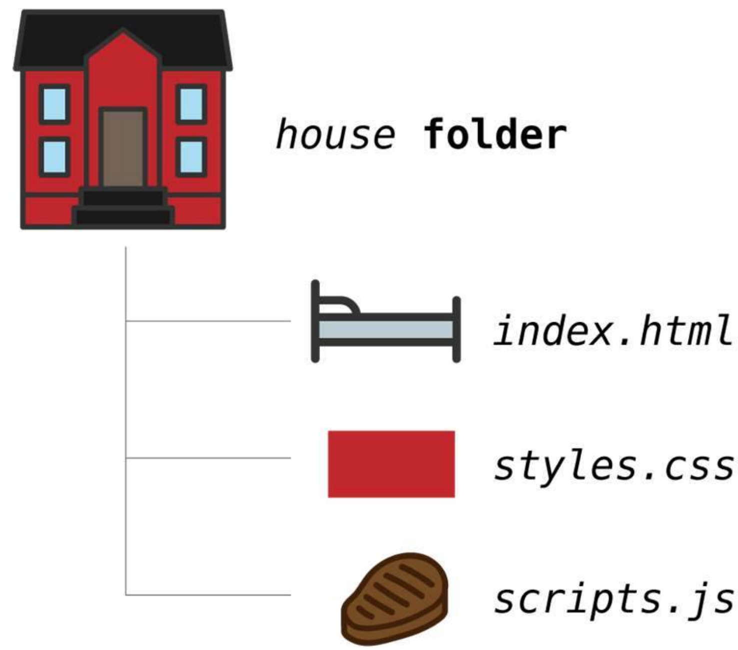


- The selector points to the HTML element you want to style.
- The declaration block contains one or more declarations separated by semicolons.
- Each declaration includes a CSS **property name** and a **value**, separated by a colon.
- A CSS declaration always ends with a semicolon, and declaration blocks are surrounded by curly braces.

ที่มา https://www.w3schools.com/css/css_syntax.asp

4. JavaScript

Html CSS JavaScript



4. JavaScript

- เดิมทีภาษา JavaScript ทำงานที่ Web browser เป็นหลัก โดยทำหน้าที่เปลี่ยนแปลงโครงสร้างของ HTML เป็นหลัก

- JavaScript Can Change HTML Content
- JavaScript Can Change HTML Attribute Values
- JavaScript Can Change HTML Styles (CSS)
- JavaScript Can Hide/Show HTML Elements

4. JavaScript

- **JavaScript Can Change HTML Content**

```
<!DOCTYPE html>
<html>
<body>

<h2>What Can JavaScript Do?</h2>

<p id="demo">JavaScript can change HTML content.</p>

<button type="button" onclick="document.getElementById('demo').innerHTML
= 'Hello JavaScript!'">Click Me!</button>

</body>
</html>
```

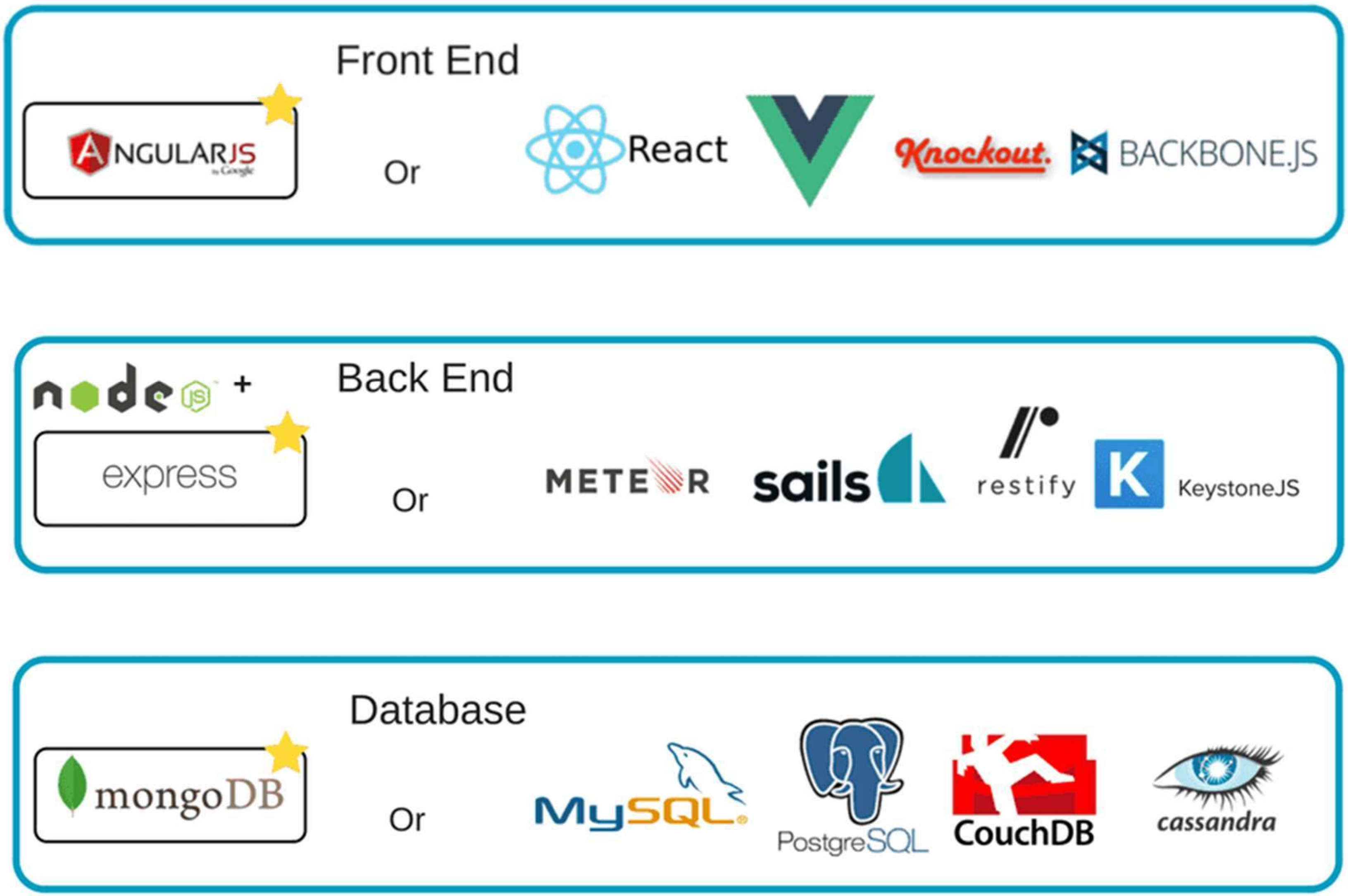
4. JavaScript

- ต่อมา ผู้คนอยากให้ JavaScript ทำงานนอก Web Browser ได้ เพื่อเพิ่มความสามารถของมัน จึงมีคนคิดค้นสร้างสิ่งที่เรียกว่า Node

- Node หรือ Node.js คือ JavaScript Runtime ที่ถูกสร้างด้วย Chrome’V8 JavaScript Engine

- โดย JavaScript Runtime ทำหน้าเพื่อ อ่าน code ภาษา java script และทำงานตามคำสั่ง เพื่อให้ os ต่างๆ ทำงานได้

Full Stack JavaScript Tools and Technologies



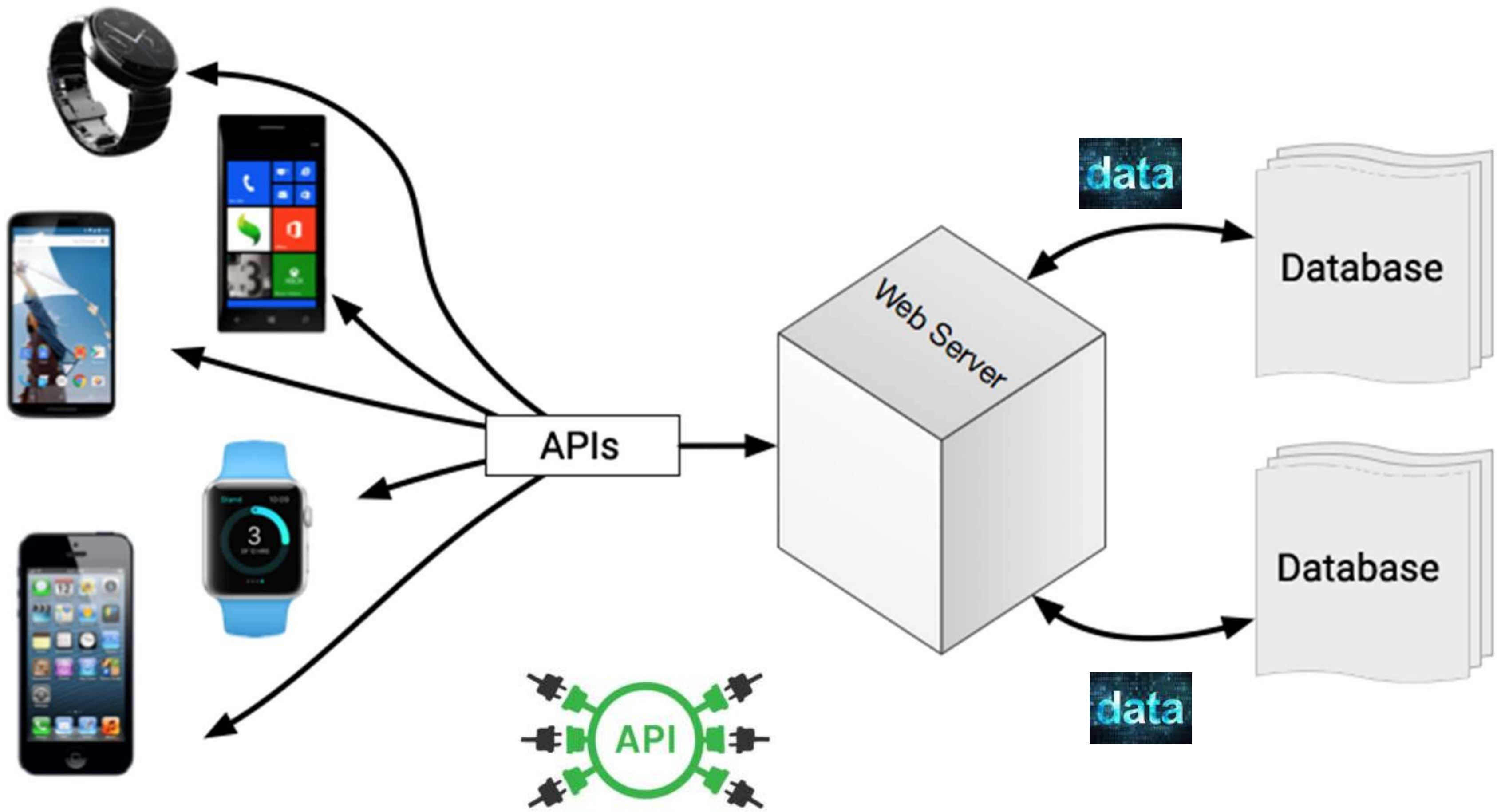
5. Introduction to API

โรงไฟฟ้า ผลิตไฟฟ้า เพื่อให้บริการไฟฟ้า

คำถาม ถ้าไม่มีปลั๊กตัวเมียมาให้ เราจะใช้ไฟฟ้า อย่างไร ?

5. Introduction to API

API ย่อมาจาก Application Programming Interface



An API allows one piece of software talk to another.

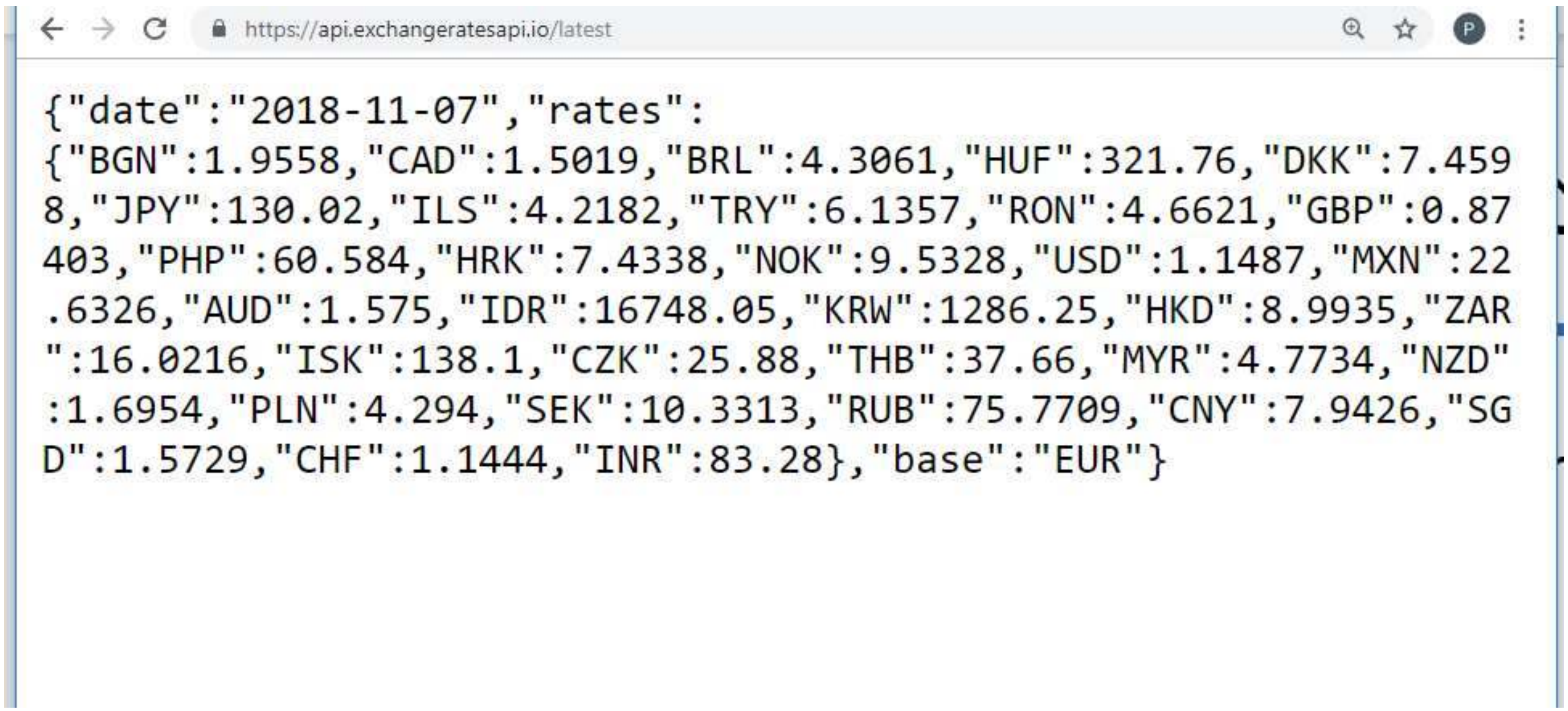
ตัวอย่าง API

Google Maps APIs

YouTube
Data API v3

ตัวอย่าง API ให้ข้อมูล อัตราแลกเปลี่ยน

https://api.exchangeratesapi.io/latest

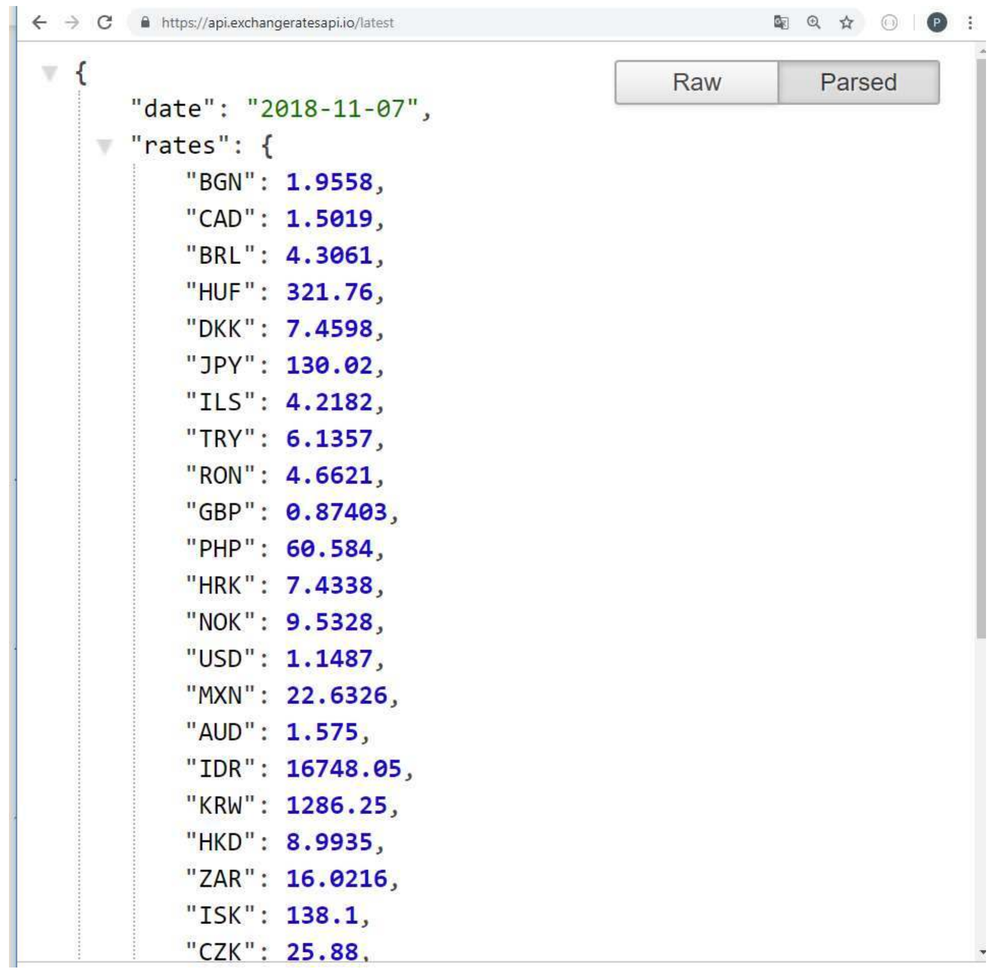

https://api.exchangeratesapi.io/latest

{"date":"2018-11-07","rates":{"BGN":1.9558,"CAD":1.5019,"BRL":4.3061,"HUF":321.76,"DKK":7.4598,"JPY":130.02,"ILS":4.2182,"TRY":6.1357,"RON":4.6621,"GBP":0.87403,"PHP":60.584,"HRK":7.4338,"NOK":9.5328,"USD":1.1487,"MXN":22.6326,"AUD":1.575,"IDR":16748.05,"KRW":1286.25,"HKD":8.9935,"ZAR":16.0216,"ISK":138.1,"CZK":25.88,"THB":37.66,"MYR":4.7734,"NZD":1.6954,"PLN":4.294,"SEK":10.3313,"RUB":75.7709,"CNY":7.9426,"SGD":1.5729,"CHF":1.1444,"INR":83.28},"base":"EUR"}

ตัวอย่าง API ให้ข้อมูล อัตราแลกเปลี่ยน

https://api.exchangeratesapi.io/latest

ติดตั้ง plugin **json formatter**



การเขียน html ดึงค่าข้อมูล ผ่าน API

https://api.exchangeratesapi.io/latest

ไฟล์ index.html

```
<html>
<head>
	<title>Welcome to JSON</title>
	<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.3.1/jquery.min.js"></script>
</head>
<body>

	<script type="text/javascript">
		$.getJSON('https://api.exchangeratesapi.io/latest', function(response) {
			document.getElementById("data").innerHTML = response.rates.THB
			document.getElementById("data1").innerHTML = response.date
		});
	</script>

	<div style = "color:red"   id="data" > </div>
	<div style = "color:blue"  id="data1"> </div>


</body>
</html>
```

Request

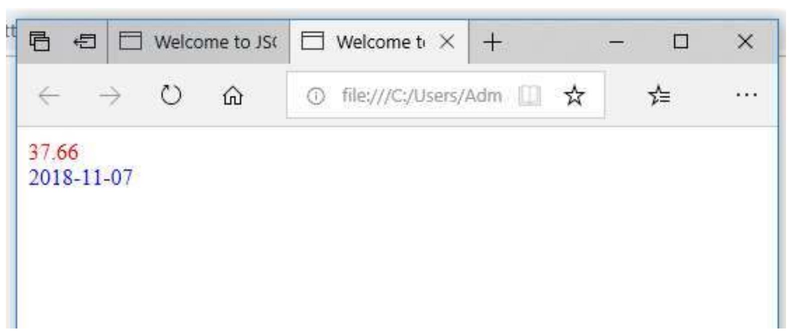
Data

https://api.exchangeratesapi.io/latest


```
"PHP": 60.584,
"HRK": 7.4338,
"NOK": 9.5328,
"USD": 1.1487,
"MXN": 22.6326,
"AUD": 1.575,
"IDR": 16748.05,
"KRW": 1286.25,
"HKD": 8.9935,
"ZAR": 16.0216,
"ISK": 138.1,
"CZK": 25.88,
"THB": 37.66,
"MYR": 4.7734,
"NZD": 1.6954,
"PLN": 4.294,
"SEK": 10.3313,
"RUB": 75.7709,
"CNY": 7.9426,
"SGD": 1.5729,
"CHF": 1.1444,
"INR": 83.28
},
"base": "EUR"
```


เปิดไฟล์ ผ่าน **Web browser**

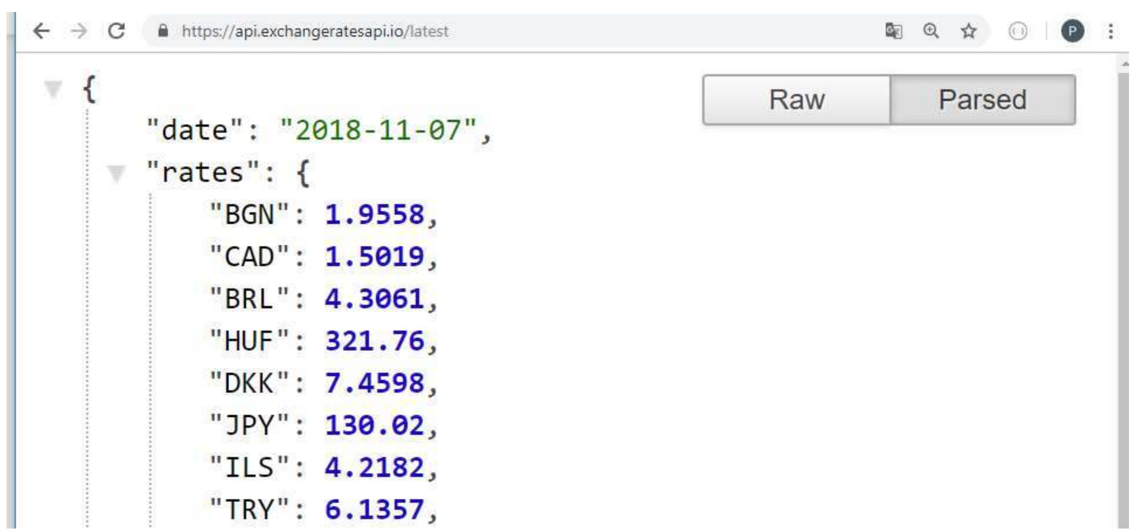
แสดงผล



REST API

- **REST** หรือ Representational State Transfer มันเป็นวิธีในการสร้าง Web Service รูปแบบนึงที่อาศัย HTTP Method (GET, POST, PUT, DELETE) ในการทำงาน และส่งผลกลับมาในรูปแบบของ JSON หรือ XML

เช่น https://api.exchangeratesapi.io/latest



REST API

ตัวอย่าง API Document โดยใช้ Swagger

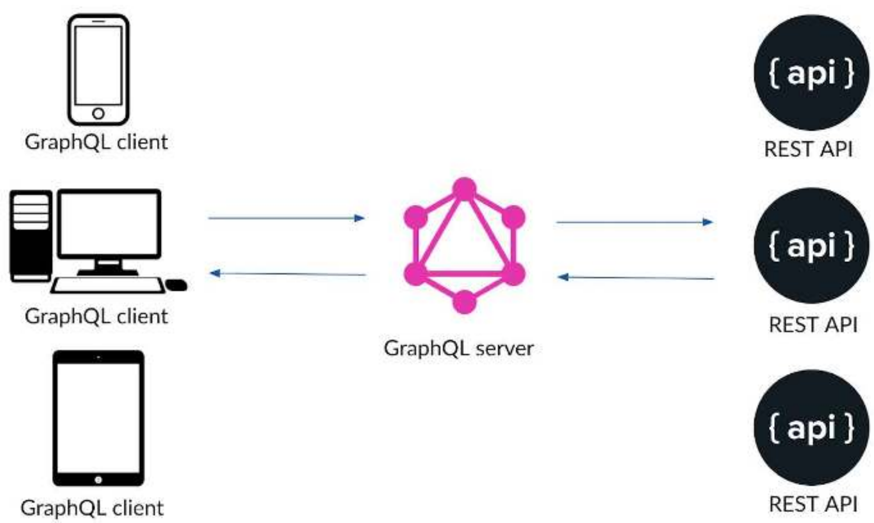
workshop$Order Entity CRUD operations

GET	`/entities/workshop$Order`	Gets a list of entities: workshop$Order
POST	`/entities/workshop$Order`	Creates new entity: workshop$Order
GET	`/entities/workshop$Order/{entityId}`	Gets a single entity by identifier: workshop$Order
PUT	`/entities/workshop$Order/{entityId}`	Updates the entity: workshop$Order
DELETE	`/entities/workshop$Order/{entityId}`	Deletes the entity: workshop$Order
GET	`/entities/workshop$Order/search`	Find entities by filter conditions: workshop$Order
POST	`/entities/workshop$Order/search`	Find entities by filter conditions: workshop$Order

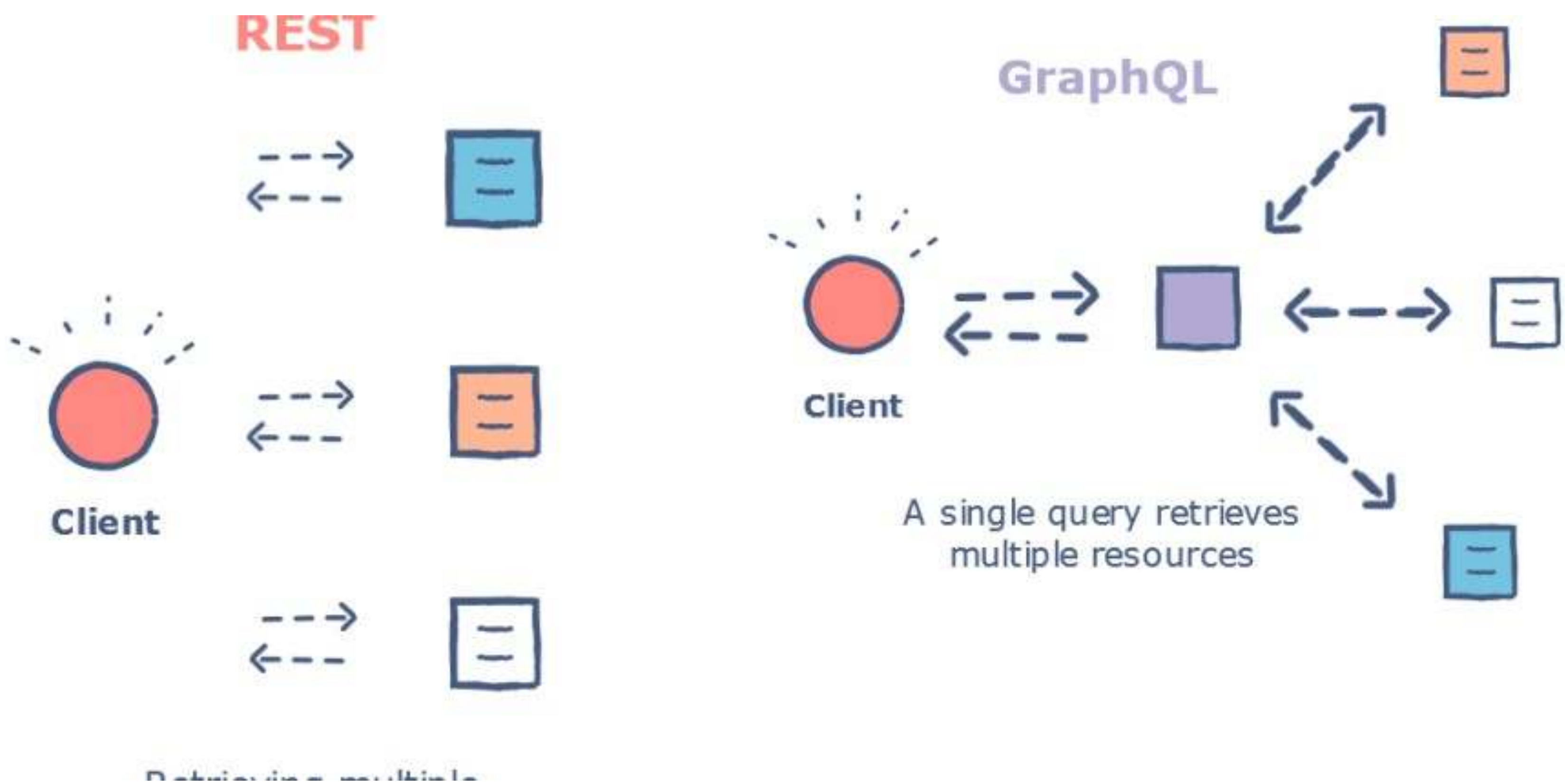
GraphQL

- เนื่องจาก การดึงค่าแบบ Rest API บางครั้งข้อมูลมากเกินความจำเป็น บางครั้งต้องดึงหลายครั้ง เพื่อนำข้อมูลมาประมวลผล หรือเอามาประกอบกัน จึงทำให้เกิดแนวคิด ของ GraphQL

“ GraphQL is a query language for APIs and a runtime for fulfilling those queries with your existing data. ”



GraphQL


REST
Client
GraphQL
Client
A single query retrieves
multiple resources

ขอบคุณครับ